



Assignment 07: Working with Sets in Python

Instructions

1. Write each program in a separate Python file.
 2. Use meaningful variable names and proper indentation.
 3. Display clear output messages.
 4. Use set operations wherever applicable.
 5. Test your programs with different values.
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◆ 1 Create and Display a Set

Problem:

Create a set containing the names of 5 fruits and display the set.

Example Output

```
Fruits = {'Apple', 'Mango', 'Orange', 'Banana', 'Grapes'}
```

+ 2 Add an Element to a Set

Problem:

Create a set of colors and add a new color using the `add()` method.

Example Output

```
Original Set = {'Red', 'Blue', 'Green'}  
Updated Set = {'Red', 'Blue', 'Green', 'Yellow'}
```

— 3 Remove an Element from a Set

Problem:

Create a set of cities and remove one city using the `remove()` method.

Example Output

```
Original Set = {'Mumbai', 'Delhi', 'Bangalore'}
```

```
Updated Set = {'Mumbai', 'Bangalore'}
```



Check if an Element Exists

Problem:

Create a set of student names and check whether a particular name exists in the set.

Example Output

```
Students = {'Ahmed', 'Fatima', 'Rahul'}
```

```
Rahul is present in the set.
```



Find the Union of Two Sets

Problem:

Create two sets and display all unique elements from both sets using the `union()` method.

Example Output

```
Set A = {1, 2, 3, 4}
```

```
Set B = {3, 4, 5, 6}
```

```
Union = {1, 2, 3, 4, 5, 6}
```



Find the Intersection of Two Sets

Problem:

Create two sets and find the common elements using the `intersection()` method.

Example Output

```
Set A = {10, 20, 30, 40}
```

```
Set B = {30, 40, 50, 60}
```

```
Intersection = {30, 40}
```



Find the Difference Between Two Sets

Problem:

Create two sets and display elements that are present in the first set but not in the second set.

Example Output

```
Set A = {1, 2, 3, 4, 5}
Set B = {4, 5, 6, 7}

Difference = {1, 2, 3}
```

**8****Remove Duplicate Values from a List****Problem:**

Given a list containing duplicate values, convert it into a set and display only unique values.

Example Output

```
List = [10, 20, 10, 30, 20, 40, 50]

Unique Values = {10, 20, 30, 40, 50}
```

**Bonus Question**

Find Subjects Taken by All Students

Problem:

Three students have selected subjects as shown below:

```
student1 = {"Math", "Science", "English"}
student2 = {"Science", "English", "Computer"}
student3 = {"English", "Science", "History"}
```

Find the subjects that are common to all three students.

Example Output

```
Common Subjects = {'Science', 'English'}
```

Learning Outcomes

After completing this assignment, students will be able to:

✓ Create and display sets ✓ Add and remove elements from sets ✓ Check membership using `in` ✓
Perform union, intersection, and difference operations ✓ Use sets to remove duplicate values ✓ Solve real-
world problems using set operations